Dynamic Network Analysis

An Introduction

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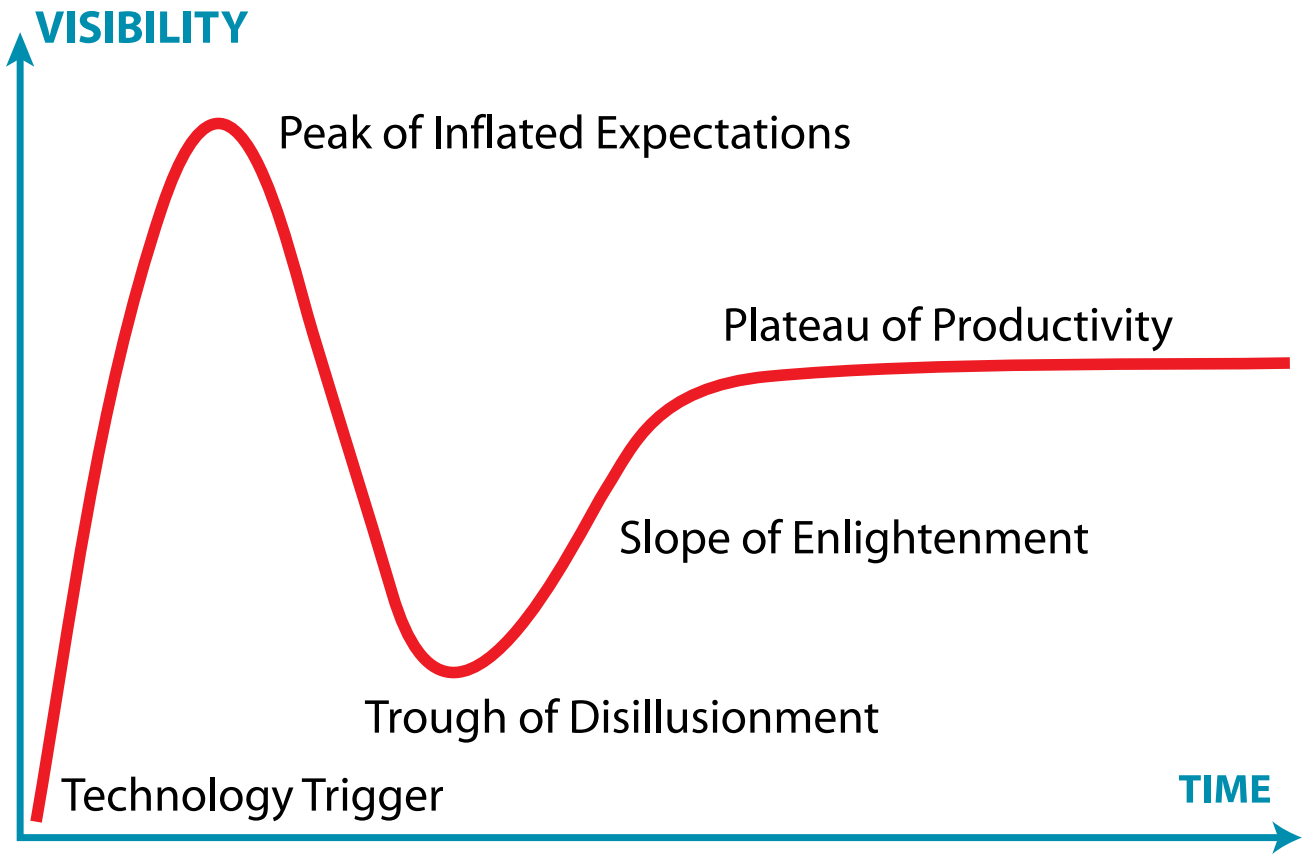
Materials and Goals

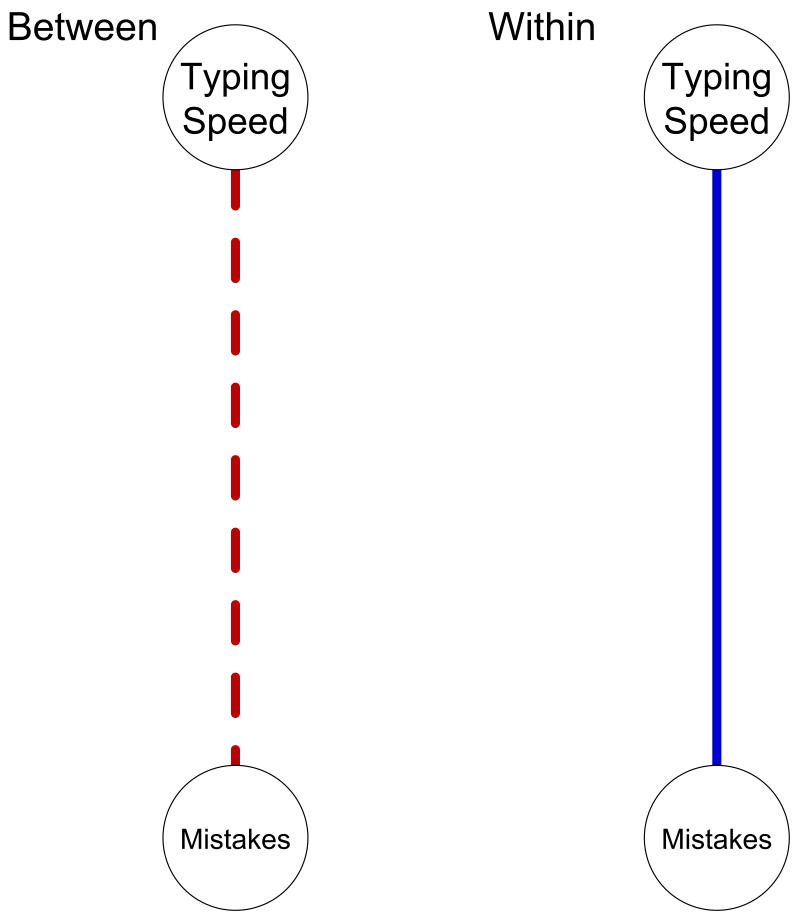
What we don’t cover

Hype Cycle



Jeremykemp, Wikimedia

Why Longitudinal Data?



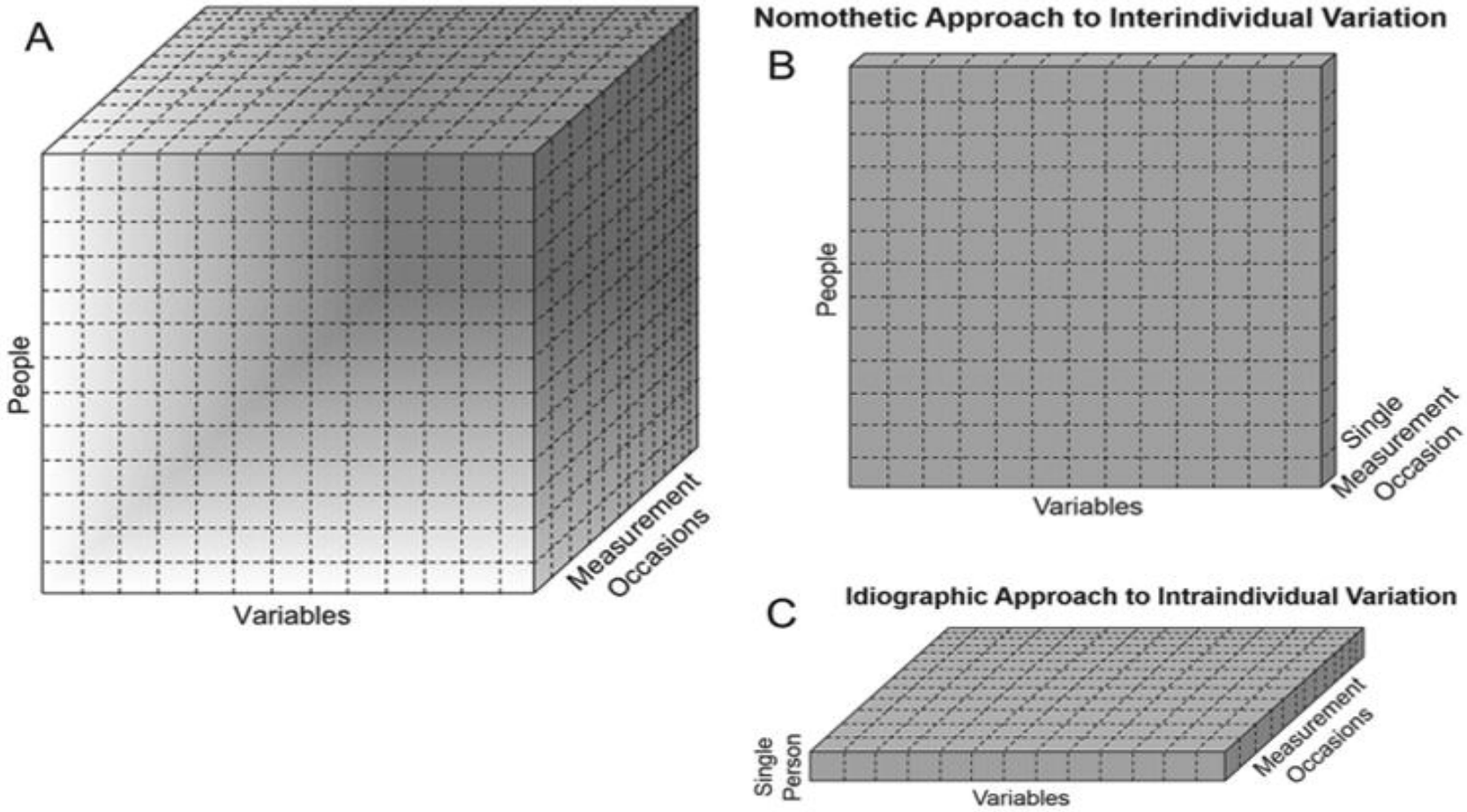
Oriented at a figure by Ellen Hamaker

Dynamic Networks

General principles

TODO ask people before what they think a dynamic network even is?

Time Series Data

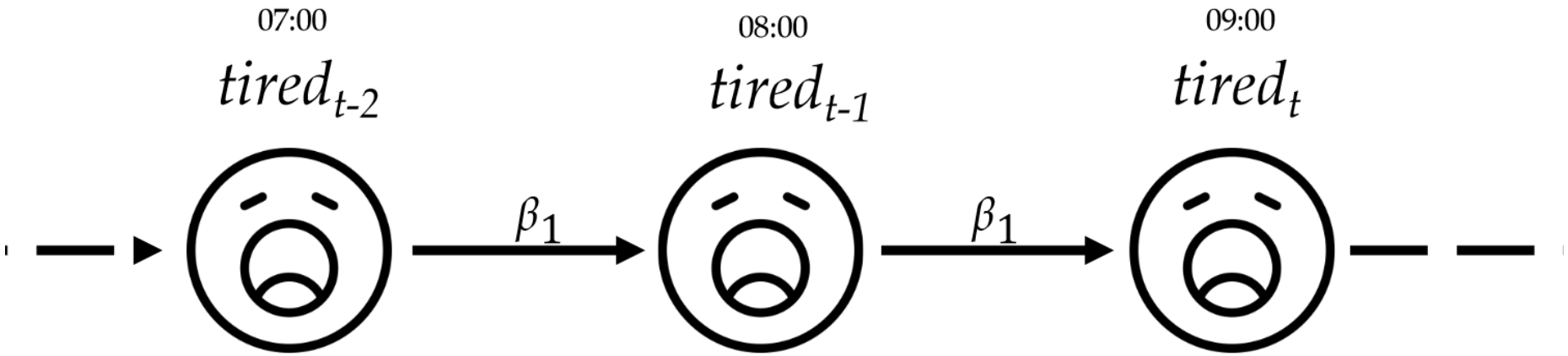


Beltz et al. (2016)

Experience Sampling

Sabina Ciesielska (Unsplash), Jonas Leupe (Unsplash)

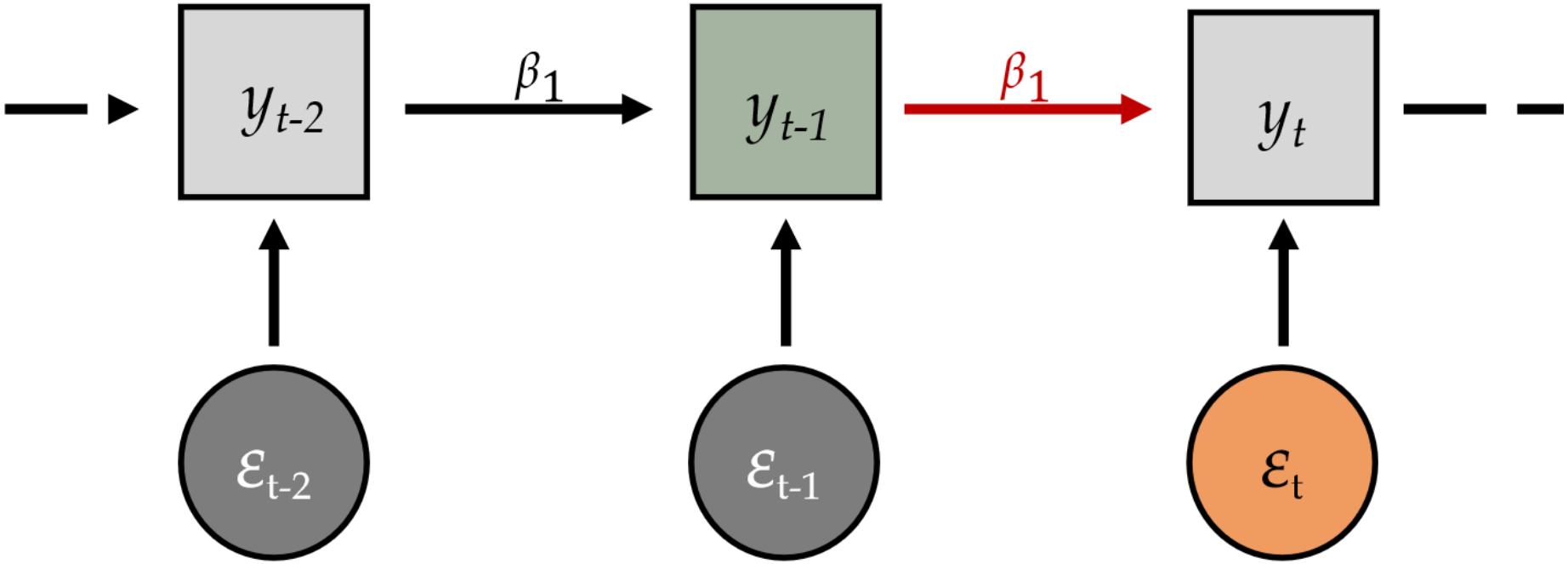
Autoregressive



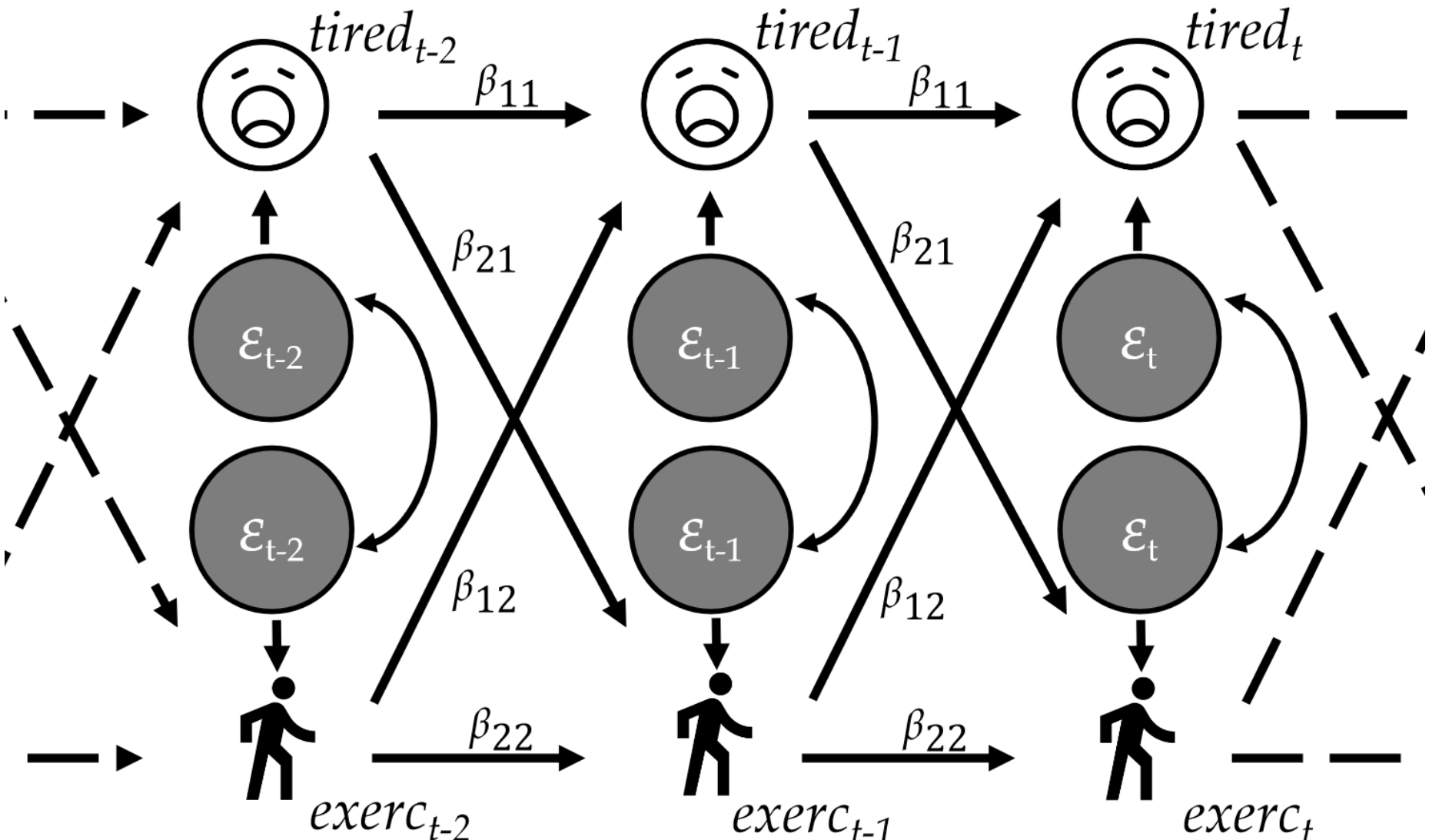
Next slides by Julian Burger (Link)

Autoregressive

$$y_t = \beta_1 y_{t-1} + \varepsilon_t$$



Cross-Lagged



Estimation of idiographic networks

Formalization

We define:

$$G = (V, E)$$

as an undirected graph consisting of nodes $V = \{1, \ldots, p\}$ and edges $E \subseteq V \times V$.

We associate a random variable X_v to each node $v \in V$. X_v takes values in space $\mathscr{X}_v$.

Theorem 1:

Maths Break

https://media.giphy.com/media/lHfxDepSGlzom6f65K/giphy.gif

Intermediate Summary

TODO change better - Attractive alternative to conceptualization of pyschopathology - Networks are everywhere - Almost any multivariate model can be a network - A lot of cross-sectional studies with limited informativeness - Longitudinal data allows for new insights

Interpreting dynamic networks

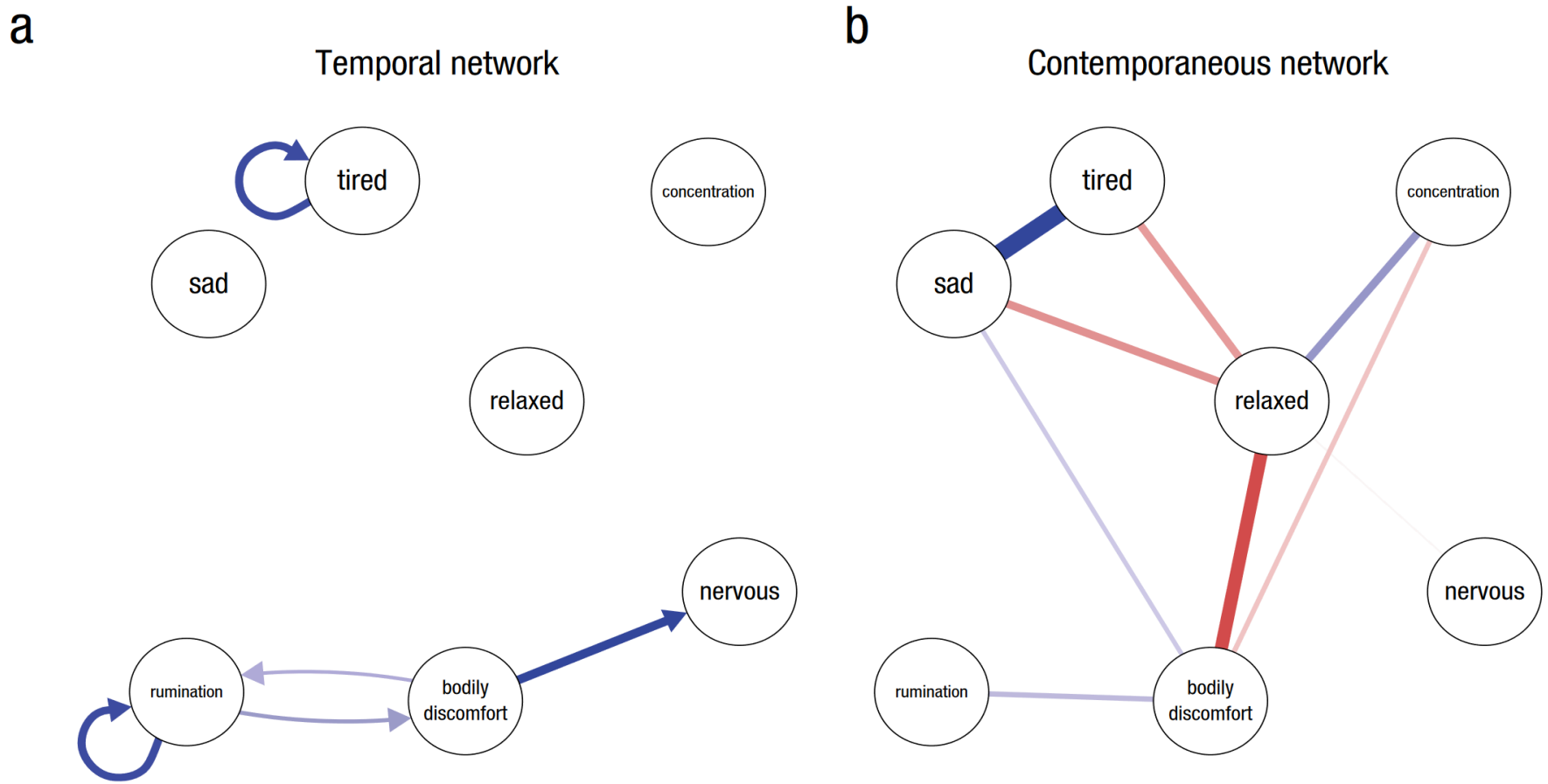
Temporal Network

“Contemporaneous” Network

TODO insert part of the image from earlier

Example from Epskamp et al. (2018)

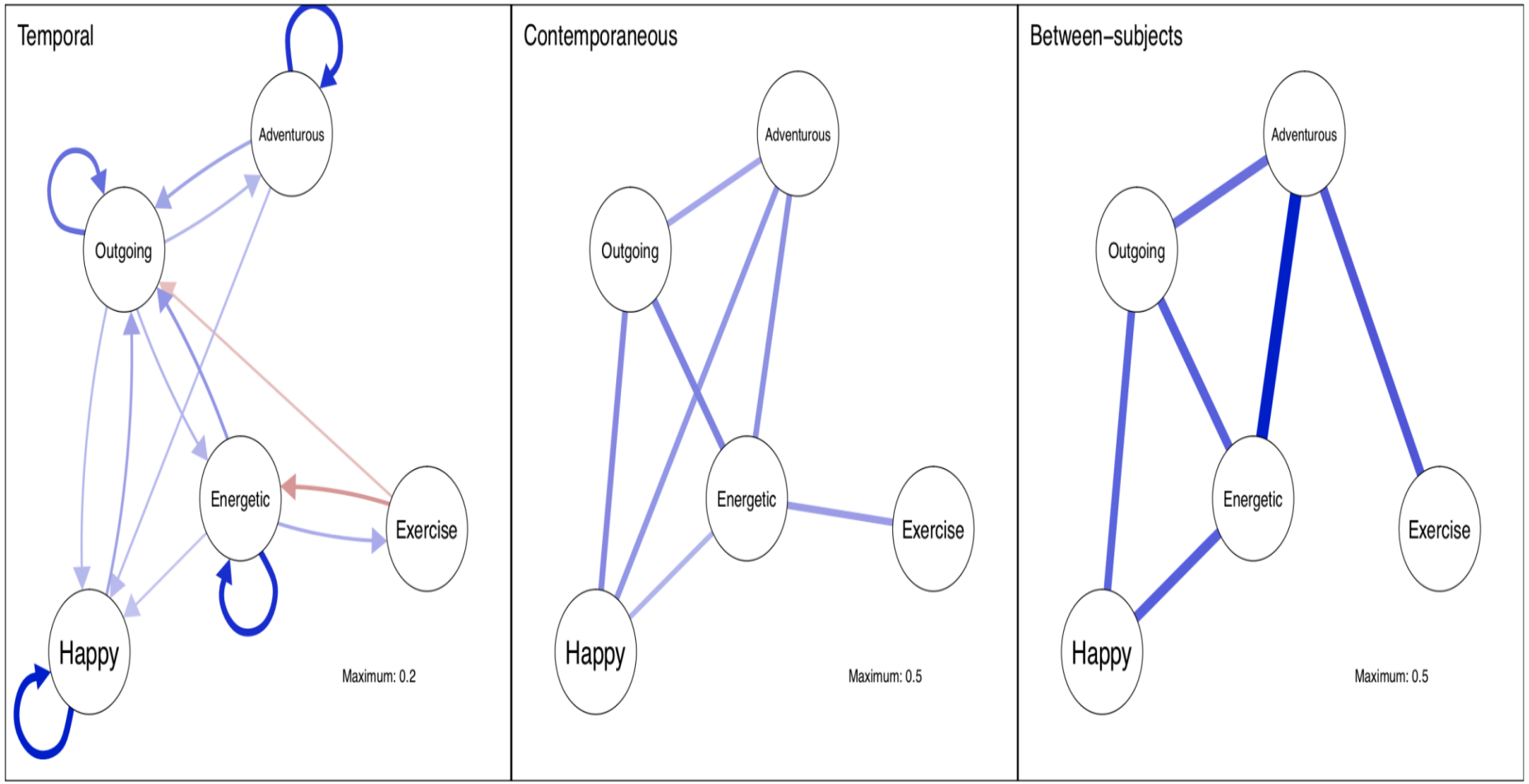
Example: Epskamp et al. (2018)



$n > 1$ Networks

Simple overview in Jordan, Winer, and Salem (2020)

Multilevel Example



Epskamp et al. (2018)

Interpretation

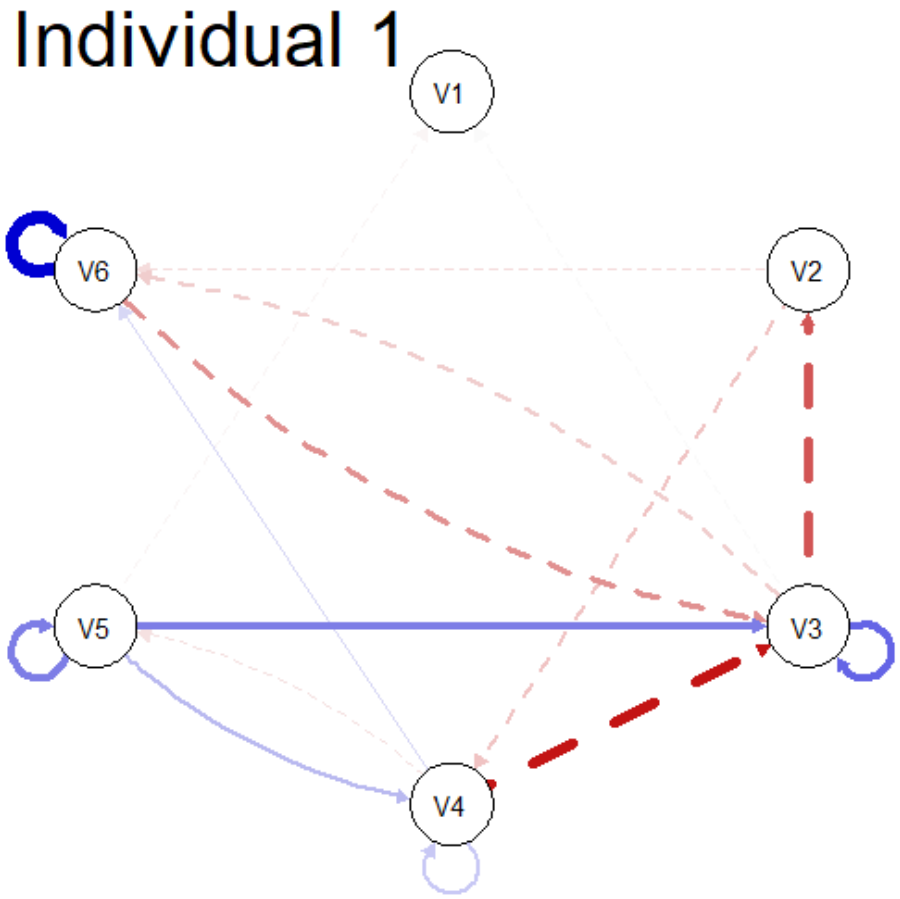
Interpretation

Implementations

Evidence and Pitfalls

Topics to be covered

Problem: Estimation Instability



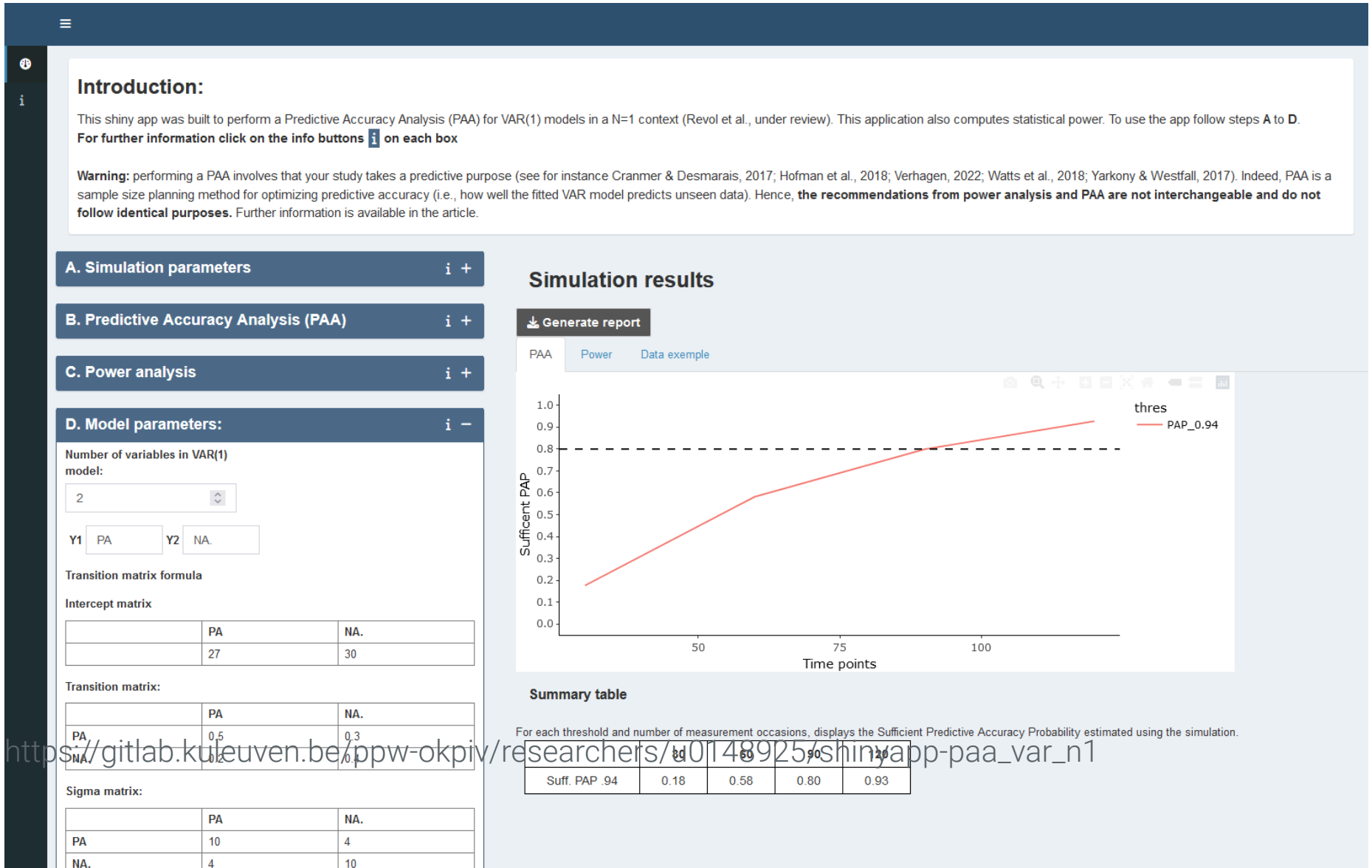
Hoekstra, Epskamp, and Borsboom (2022)

Feasibility of Idiographic Networks

Mansueto et al. (2023), Siepe et al. (2024)

Solutions for Estimation Instability

Solutions for Estimation Instability

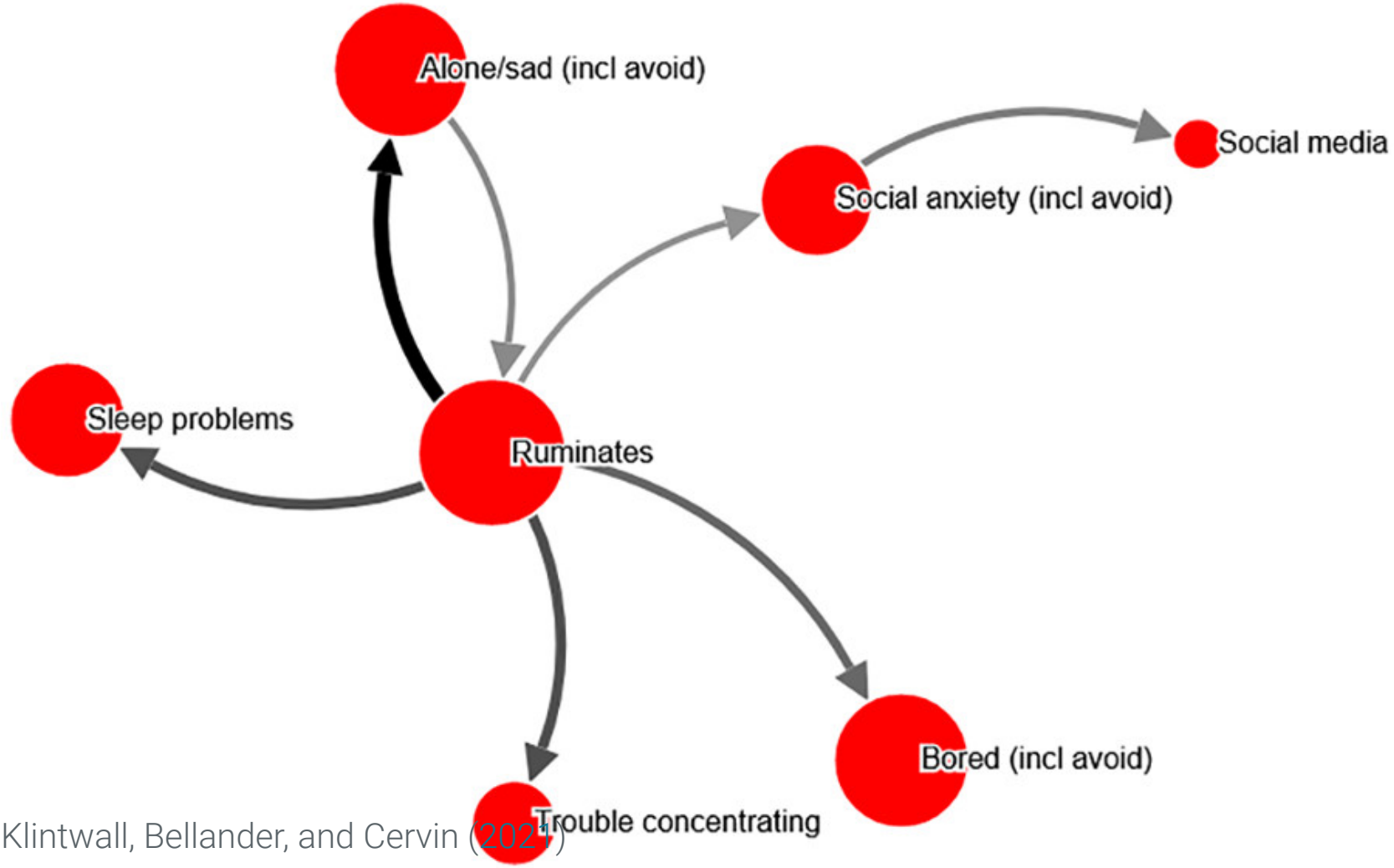


https://gitlab.kuleuven.be/ppw-okpiv/researchers/u0148925/shinyapp-paa_var_n1

Evidence in Clinical Practice

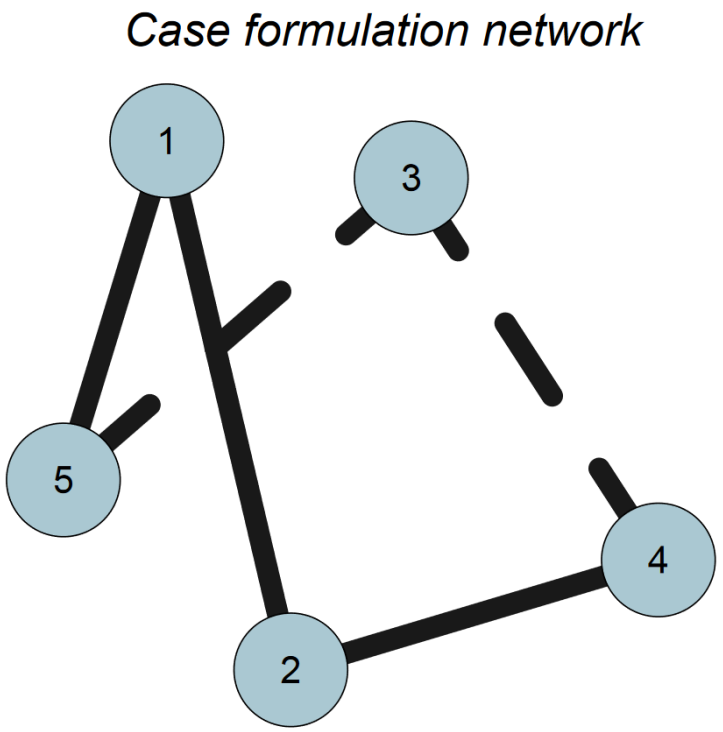
Recent developments

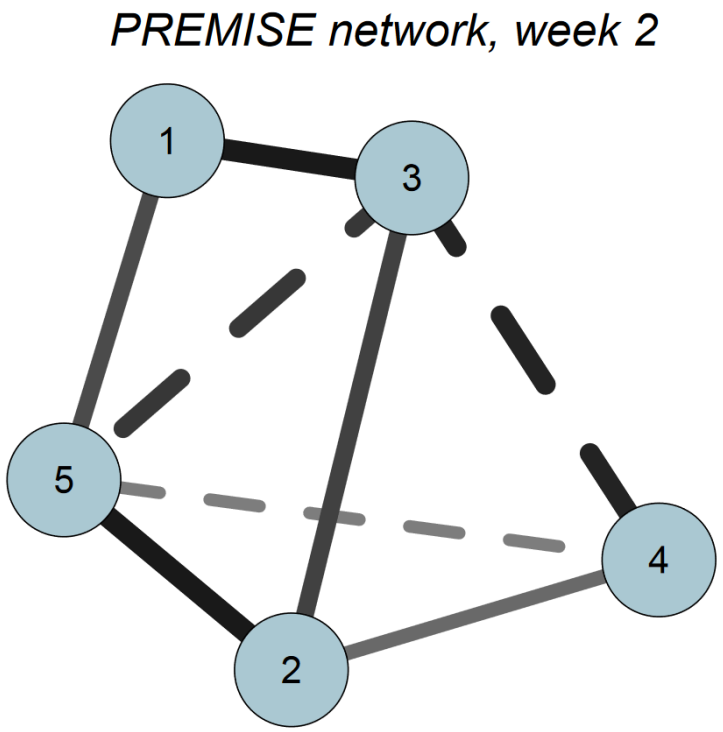
PECAN



Klintwall, Bellander, and Cervin (2021)

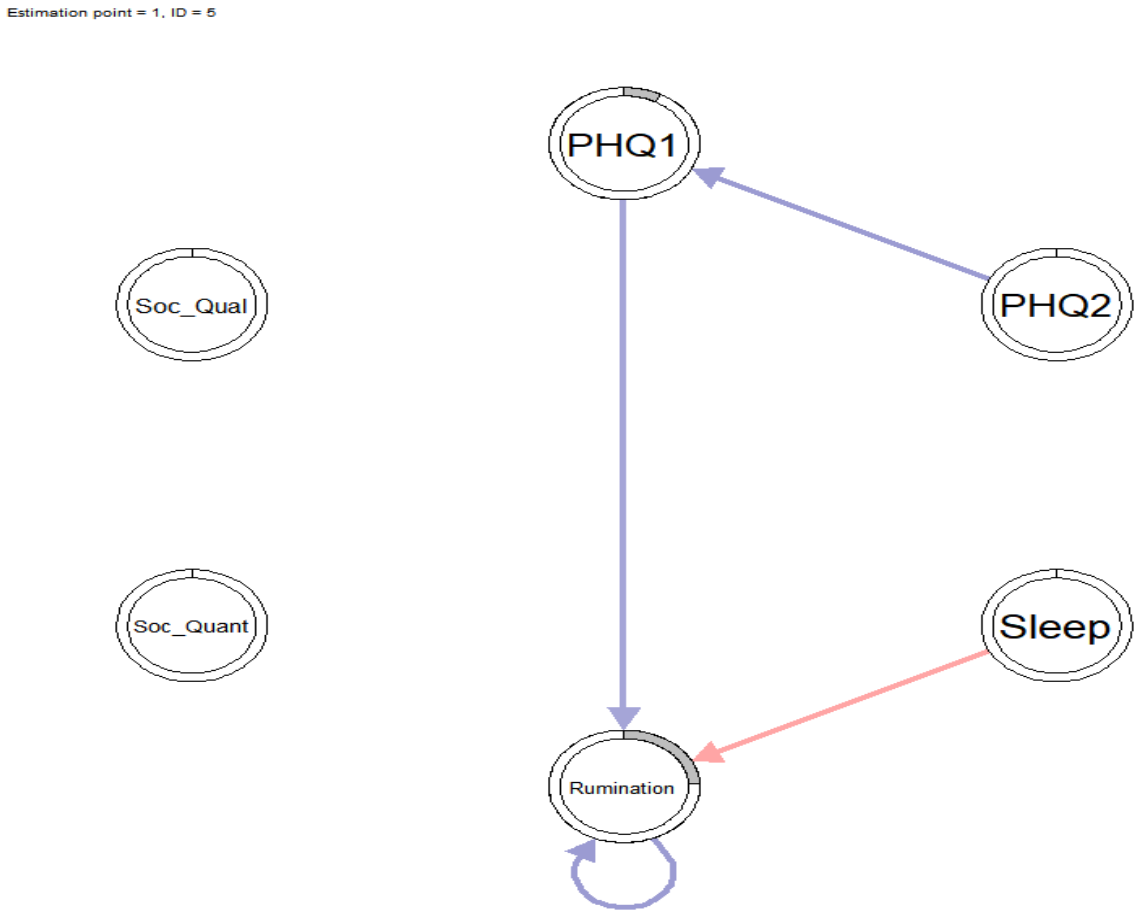
PREMISE





Burger et al. (2022), Burger et al. (2024), Björn S. Siepe, Kloft, and Heck (2024)

Time varying networks



Siepe et al. (2024), Haslbeck, Bringmann, and Waldorp (2021), Scholten et al. (2023)

Prediction

TODO add conceptual introduction, which information do we use? also add that we could use our network itself to predict, people should try that more often.

Prediction

Prediction

Using network analysis for the prediction of treatment dropout in patients with mood and anxiety disorders: A methodological proof-of-concept study

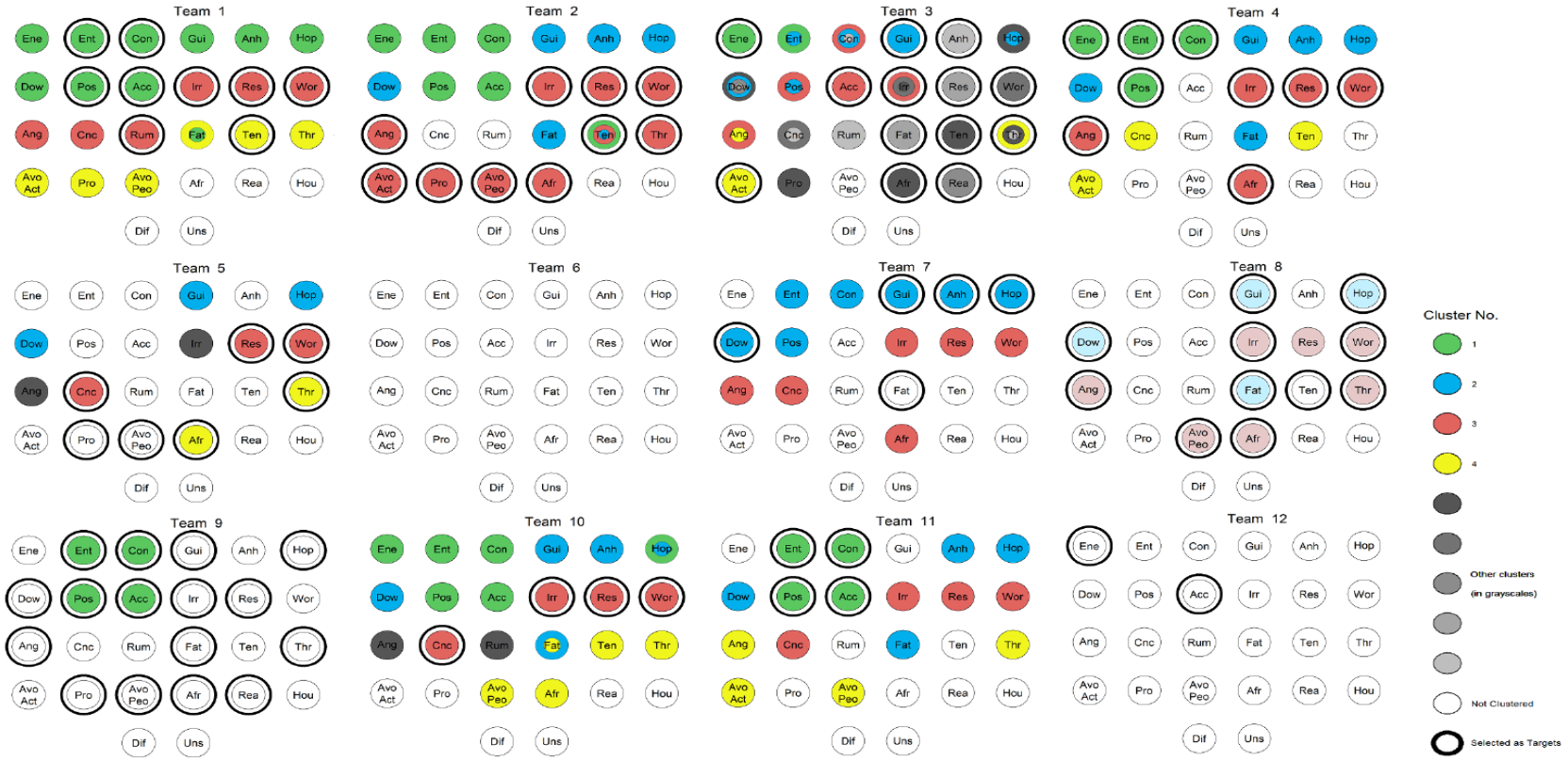
Wolfgang Lutz[1], Brian Schwartz[1], Stefan G. Hofmann[2], Aaron J. Fisher[3], Kristin Husen[1] & Julian A. Rubel[1]

There are large health, societal, and economic costs associated with attrition from psychological services. The recently emerged, innovative statistical tool of complex network analysis was used in the present proof-of-concept study to improve the prediction of attrition. Fifty-eight patients undergoing psychological treatment for mood or anxiety disorders were assessed using Ecological Momentary Assessments four times a day for two weeks before treatment (3,248 measurements). Multilevel vector autoregressive models were employed to compute dynamic symptom networks. Intake variables and network parameters (centrality measures) were used as predictors for dropout using machine-learning algorithms. Networks for patients differed significantly between completers and dropouts. Among intake variables, initial impairment and sex predicted dropout explaining 6% of the variance. The network analysis identified four additional predictors: Expected force of being excited, outstrength of experiencing social support, betweenness of feeling nervous, and instrength of being active. The final model with the two intake and four network variables explained 32% of variance in dropout and identified 47 out of 58 patients correctly. The findings indicate that patients' dynamic network structures may improve the prediction of dropout. When implemented in routine care, such prediction models could identify patients at risk for attrition and inform personalized treatment recommendations.

Lutz et al. (2018)

Prediction

Which analyses?



Bastiaansen et al. (2020)

Analytic uncertainty: Go multiverse

include my GIMME paper here

Björn S. Siepe and Heck (2023)

Summary

Outlook

Main Takeaways

Get in Touch

Feel

free

to

contact

me

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Resources

Materials

PNAWS 2020 - freely accessible. Partly served as inspiration for this workshop, especially the slides by Julian Burger.

Psych Networks Blog - no longer active, but interesting backlog

Workshops by Noémi Schuurman - multiple materials for multilevel models in R and MPlus

Applied Workshop by Me - slides including more code examples etc.

Figures

Jeremykemp at English Wikipedia (https://commons.wikimedia.org/wiki/File:Gartner_Hype_Cycle.svg), „Gartner Hype Cycle", https://creativecommons.org/licenses/by-sa/3.0/legalcode

Foto von Sabina auf Unsplash

Foto von Jonas Leupe auf Unsplash

Rumination icons created by Freepik - Flaticon

References

Bastiaansen, Jojanneke A., Yoram K. Kunkels, Frank J. Blaauw, Steven M. Boker, Eva Ceulemans, Meng Chen, Sy-Miin Chow, et al. 2020. “Time to Get Personal? The Impact of Researchers Choices on the Selection of Treatment Targets Using the Experience Sampling Methodology.” *Journal of Psychosomatic Research* 137 (October): 110211. https://doi.org/10.1016/j.jpsychores.2020.110211.

Beltz, Adriene M., Aidan G. C. Wright, Briana N. Sprague, and Peter C. M. Molenaar. 2016. “Bridging the Nomothetic and Idiographic Approaches to the Analysis of Clinical Data.” *Assessment* 23 (4): 447–58. https://doi.org/10.1177/1073191116648209.

Berkhout, Sophie W., Noémi Katalin Schuurman, and Ellen Hamaker. 2024. “Let Sleeping Dogs Lie? How to Deal with the Night Gap Problem in ESM Data.” May 28, 2024. https://doi.org/10.31234/osf.io/2hec8.

Bringmann, Laura F. 2024. “The Future of Dynamic Networks in Research and Clinical Practice.” *World Psychiatry* 23 (2): 288–89. https://doi.org/10.1002/wps.21209.

Burger, Julian, Vida Andikkhash, Nelly Jäger, Therese Anderbro, Tessa F. Blanken, and Lars Klintwall. 2024. “A Novel Approach for Constructing Personalized Networks from Longitudinal Perceived Causal Relations.” *Behaviour Research and Therapy* 173 (February): 104456. https://doi.org/10.1016/j.brat.2023.104456.

Burger, Julian, Sacha Epskamp, Date C. van der Veen, Fabian Dablander, Robert A. Schoevers, Eiko I. Fried, and Harriëtte Riese. 2022. “A Clinical PREMISE for Personalized Models: Toward a Formal Integration of Case Formulations and Statistical Networks.” *Journal of Psychopathology and Clinical Science* 131 (8): 906–16. https://doi.org/10.1037/abn0000779.